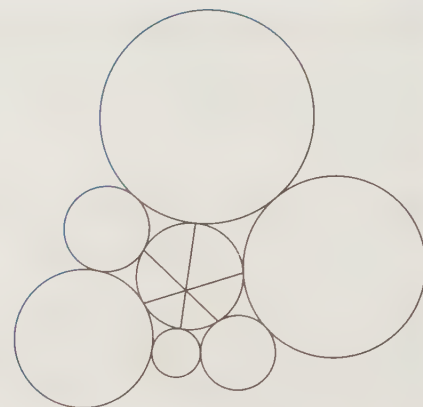




The Seven Circles Theorem

Everything in nature adheres to the cone, the cylinder and the cube. – Paul Cezanne

CHAPTER 15

Curved Surfaces

15.1 Cylinders

Instead of using congruent polygons for bases as we do with a prism, suppose we use congruent circles. The resulting solid is called a **cylinder**, an example of which is shown below. The line connecting the centers of the bases is called the **axis** of the cylinder ($\overline{OO'}$ in the diagram) and the radius of a base is also considered the radius of the cylinder.

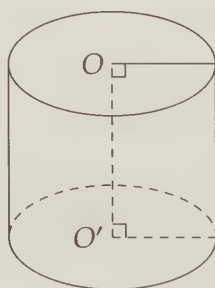


Figure 15.1: A Right Circular Cylinder

You won't be surprised to learn that if the axis is perpendicular to the bases, then the cylinder is a **right cylinder** as shown above. Furthermore, we can have cylinders in which the bases are any sort of wacky curves. Therefore, our simple cylinder above is more precisely called a **right circular cylinder**, where the 'circular' tells us that the bases are circles. When we speak of a cylinder, we nearly always

mean a 'right circular cylinder,' instead of one of the wacky cylinders shown in Figure 15.2 below.

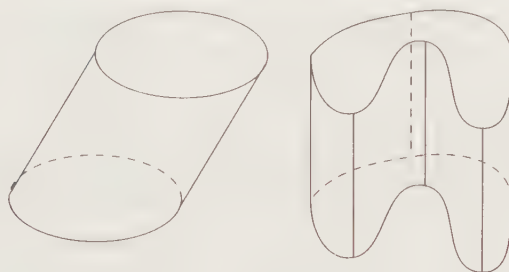


Figure 15.2: More Cylinders

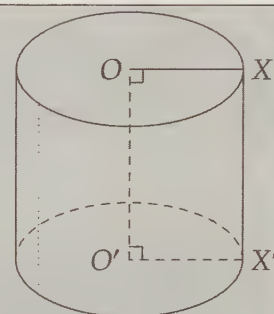
Problems

Problem 15.1:

- Suppose we take a cross-section of a cylinder that is perpendicular to the axis of the cylinder. What shape is this cross-section?
- What is the shape of a cross-section of a cylinder that contains the axis of the cylinder?
- What is the shape of a cross-section of a cylinder that is parallel to the axis of the cylinder?

Problem 15.2: The figure at right shows a right circular cylinder (a.k.a. a cylinder) with radius 3 and height 5.

- Find the volume of the cylinder.
- Suppose we cut the curved surface of the cylinder along the dotted line shown (assume the dotted line is perpendicular to the bases) and 'unroll' the surface into a plane. What type of figure results?
- What is the lateral surface area of the cylinder?
- What is the total surface area of the cylinder?
- Find formulas for the volume, lateral surface area, and total surface area of a cylinder with radius r and height h .



Problem 15.3: The radius of a cylinder is $\frac{2}{3}$ its height. Find the total surface area of the cylinder if its volume is 96π .

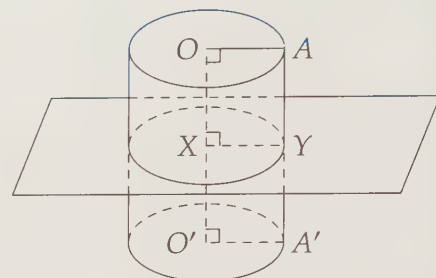
We'll start our investigation of cylinders by investigating a few cross-sections of a cylinder.

Problem 15.1:

- Suppose we take a cross-section of a cylinder that is perpendicular to the axis of the cylinder. What shape is this cross-section?
- What is the shape of a cross-section of a cylinder that contains the axis of the cylinder?
- What is the shape of a cross-section of a cylinder that is parallel to the axis of the cylinder?

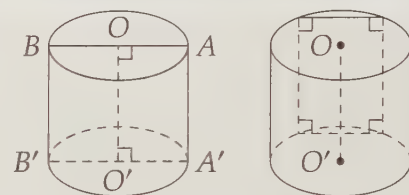
Solution for Problem 15.1:

- (a) Intuitively, it seems clear that the cross-section is a circle. To prove this, we consider A and A' on the bases such that $\overline{AA'}$ is parallel to the axis $\overline{OO'}$. We let X and Y be the points where $\overline{OO'}$ and $\overline{AA'}$, respectively, meet our cross-section plane. Since this plane is perpendicular to the axis, we have $\overline{XY} \perp \overline{OO'}$. So, $XYAO$ is a rectangle, and $XY = OA$. Hence, XY equals the radius of the cylinder.



But there's nothing special about the points A and A' we chose on the circumferences of the bases! In exactly the same way, we can show that any point on the intersection of our plane and the curved surface of the cylinder is exactly the cylinder's radius away from point X on the axis. Therefore, our cross-section is indeed a circle.

- (b) Since $\overline{AA'}$ and $\overline{BB'}$ are both parallel to the axis of the cylinder, they are perpendicular to both bases. So, $AA'B'B$ is a rectangle.
- (c) Since our cross-section is parallel to the axis of the cylinder, it must be perpendicular to both bases. Therefore, as in the previous part, our cross-section is a rectangle.



□

One result of part (a) above is that the set of all points that are a fixed distance from a given line in space is a cylindrical surface, as shown below. The surface is basically a cylinder without bases that continues forever along the line that is its axis.

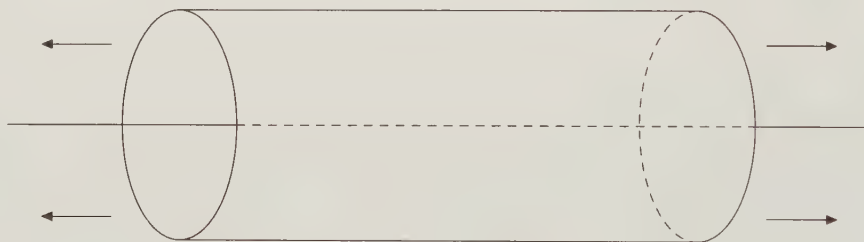
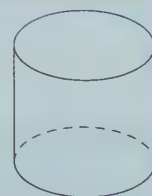


Figure 15.3: A Cylindrical Surface and Its Axis

Now that we have some sense for what a cylinder is like, let's investigate the area and volume of a cylinder.

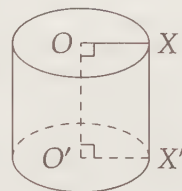
Problem 15.2: The figure shows a right circular cylinder (a.k.a. a cylinder) with radius 3 and height 5.

- (a) Find the volume of the cylinder.
- (b) What is the lateral surface area of our cylinder?
- (c) What is the total surface area of our cylinder?
- (d) Find formulas for the volume, lateral surface area, and total surface area of a cylinder with radius r and height h .

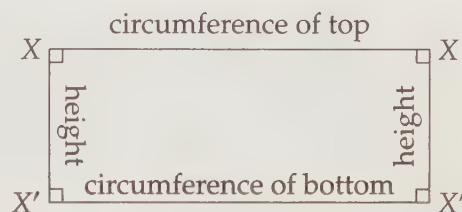


Solution for Problem 15.2:

- (a) Our cylinder is just like a prism – its cross-sections have the same area from bottom to top. Therefore, its volume is just the area of a base times the height. The area of the base is $\pi(3^2) = 9\pi$, so the volume is $(9\pi)(5) = 45\pi$.
- (b) We don't have any tools to deal with finding the area of a curved surface, so we'll have to do something clever. Suppose our cylindrical surface is made of paper, so we can cut along a segment like $\overline{XX'}$ that is parallel to both bases (and hence the same length as a height). Then we 'unroll' the surface. The top and the bottom are always 5 units apart (the height of the cylinder), so the unrolled surface is a rectangle.



The unrolled surface is shown at right. One dimension of this rectangle is the height of the cylinder; the other is the circumference of the cylinder. For our given cylinder, the height is 5 and the circumference is $2\pi r = 6\pi$. Therefore, our lateral surface area is $(5)(6\pi) = 30\pi$.



- (c) For the total surface area, we add the bases to our lateral surface area. Each base has area 9π , so the total surface area is

$$30\pi + 2(9\pi) = 48\pi.$$

- (d) We simply follow the steps we used to solve the preceding parts. The area of each base is πr^2 , so the volume is $\pi r^2 h$. The curved surface unrolls to form a rectangle with dimensions h and $2\pi r$ (the circumference of each base), so it has area $2\pi r h$. Therefore, our lateral surface area is $2\pi r h$ and the total surface area is $2\pi r h + 2\pi r^2$.

□

Important: A cylinder with height h and radius r has:



$$\begin{aligned}\text{Volume} &= \pi r^2 h \\ \text{Lateral Surface Area} &= 2\pi r h \\ \text{Total Surface Area} &= 2\pi r h + 2\pi r^2\end{aligned}$$

Don't memorize these formulas! If you take the time to understand them, they'll always be obvious to you.

Concept: Problems involving the curved surface of a cylinder can often be untangled by 'unrolling' the curved surface into a rectangle.



Let's try applying these formulas.

Problem 15.3: The radius of a cylinder is $2/3$ its height. Find the total surface area of the cylinder if its volume is 96π .

Solution for Problem 15.3: We let the height be h , so the radius is $r = 2h/3$. Since the volume is 96π , we have $\pi r^2 h = 96\pi$. Substituting for r gives

$$\left(\frac{2h}{3}\right)^2 h\pi = 96\pi,$$

and solving this equation for h gives $h = 6$. Therefore, $r = 2h/3 = 4$. Now we can find our total surface area:

$$2\pi rh + 2\pi r^2 = 48\pi + 32\pi = 80\pi.$$

□

Concept: These basic area/volume word problems are often no different from other word problems. The key to solving them is to assign variables and use the area and volume information to set up equations with the given information.

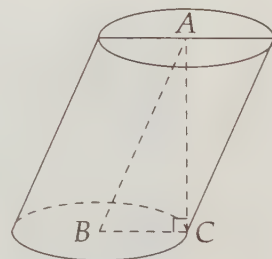
Exercises

15.1.1 A cylinder has radius 8 and height 4.

- Find the volume of the cylinder.
- Find the lateral surface area of the cylinder.
- Find the total surface area of the cylinder.

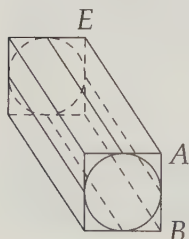
15.1.2 The diameter of a cylinder equals its height. The total surface area of the cylinder is 150π . What is the volume of the cylinder?

15.1.3 Shown at right is a circular cylinder that is *not* right. The bases have centers A and B as shown, and point C is on the circumference of a base. Given that $AB = 8$, $BC = 4$, and $\angle ACB = 90^\circ$, find the following:



- The height of the cylinder. (Note that $\overline{AB}$ is not an altitude of the cylinder because it is not perpendicular to the bases! Remember, the height of a cylinder or prism is the distance between its bases.)
- The volume of the cylinder.

15.1.4 The lateral surface area of cylinder C equals the sum of the areas of its bases. What is the ratio of the radius of C to the height of C ?

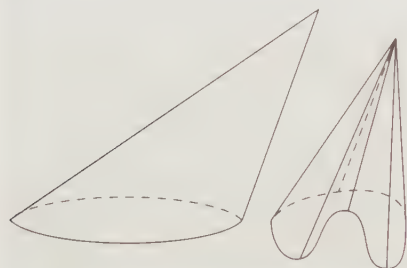
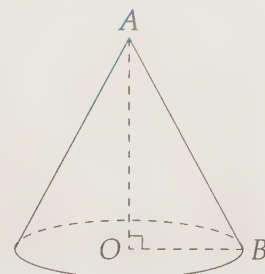


15.1.5 A cylinder is inscribed in a square prism. As shown at left, each of the bases of the cylinder is inscribed in a face of the prism, and the cylinder is tangent to the other four faces of the prism. Given that $AB = 4$ and $AE = 8$, find the volume of the cylinder.

15.1.6★ Does there exist a group of seven cylinders such that it is possible to arrange them so that each cylinder is tangent (i.e. touches at only one point) to the other six?

15.2 Cones

Just as turning the bases of a prism into circles makes a cylinder, we can turn the base of a pyramid into a circle to get a **cone**. The figure on the right shows a **right circular cone**. The point A at the tip of the cone is the **vertex** of the cone and the distance from the vertex to the base is the **height**. The line connecting the vertex to the center of the base is the **axis** of the cone. The radius of the base is considered the radius of the cone, and for right circular cones, the distance from the vertex to a point on the circumference of the base is the **slant height**.



As with cylinders, cones don't have to be 'right' or 'circular.' On the left are a couple of these less common cones. Despite the existence of these weird cones, when we simply say 'cone,' we almost always mean 'right circular cone.'

Having already learned that the volume of a pyramid equals one-third its height times the area of its base, you won't be surprised to learn that the same holds for a cone.

Important: The volume of a circular cone with height h and radius r is



$$\text{Volume} = \frac{1}{3}\pi r^2 h.$$

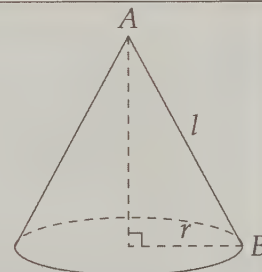
Problems

Problem 15.4:

- What is the shape of a cross-section of a cone that contains the axis of the cone?
- What is the shape of a cross-section of a cone that is perpendicular to the axis of the cone?

Problem 15.5: In this problem we will find a formula for the lateral surface area of a cone with base radius r and slant height l .

- Use the result of Problem 14.9 to take a guess at the formula.
- Let A be the vertex of the cone and B a point on the circumference of the cone. Suppose we cut the curved surface of the cone along $\overline{AB}$, then 'unroll' it into a flat figure. What type of figure do we thus form?
- How long is the curved portion of the figure you make in (b)? Use this information to find a formula for the lateral surface area of the cone.



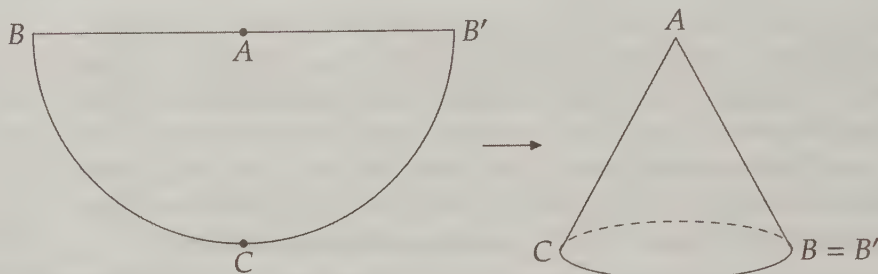
Problem 15.6: Find the volume of a right circular cone that has radius 6 and slant height 8.

Extra! *The open secret of real success is to throw your whole personality at a problem.*



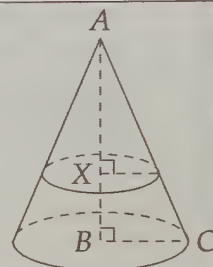
—George Polya

Problem 15.7: A semicircle is rolled up to form a cone as shown. If the radius of the semicircle is 8, what is the volume of the cone?



Problem 15.8: A cone with vertex A, height $AB = 9$, and radius $BC = 12$ is given. The cone is cut in two pieces by a plane perpendicular to $\overline{AB}$ at point X, where $AX = 6$.

- The top piece, of course, is a cone. What is the height of this cone? What is the volume of this cone?
- What is the volume of the other piece?
- What is the ratio of the volume of the little cone to the volume of the large cone? Is this surprising?



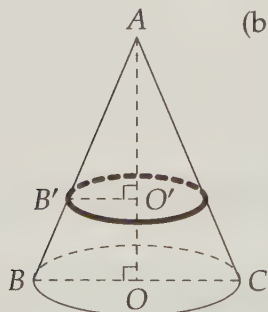
We start by getting a feel for cones by looking at some cross-sections.

Problem 15.4:

- What is the shape of a cross-section of a cone that contains the axis of the cone?
- What is the shape of a cross-section of a cone that is perpendicular to the axis of the cone?

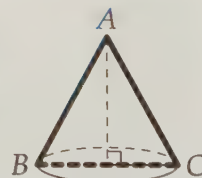
Solution for Problem 15.4:

- A cross-section of a cone that contains the cone's axis consists of two segments of equal length connecting the vertex to two points on the circumference of the base of the cone, as well as the segment connecting these two points along the base of the cone. So, our cross-section is an isosceles triangle, such as $\triangle ABC$ in the diagram at right.



- Intuitively, it seems clear that a cross-section of a cone perpendicular to the axis of the cone is a circle. Suppose the plane of our cross-section meets the axis at O' . To prove our cross-section is a circle, we must show that every point where our plane hits the curved surface of the cone is equidistant from O' . Consider point B' , the intersection of $\overline{AB}$ and our cross-section plane, as shown. Since $\overline{B'O'}$ and $\overline{BO}$ are each perpendicular to $\overline{AO}$, we have $\overline{B'O'} \parallel \overline{BO}$. Therefore, $\triangle AO'B' \sim \triangle AOB$.

Since $\triangle AO'B' \sim \triangle AOB$, we have $B'O'/BO = AO'/AO$. Therefore, we find $B'O' = (AO'/AO)(BO)$. Since BO is just the radius of the cone and AO



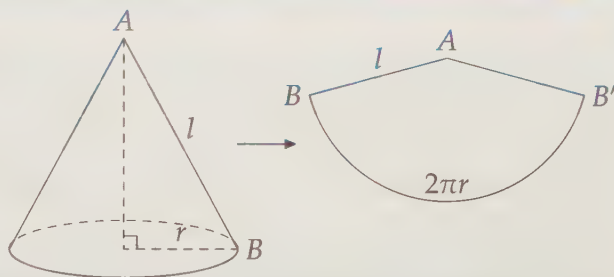
is the cone's height, we have $B'O' = (r/h)(AO')$. Similarly, we can show that all points of the cross-section are $(r/h)(AO')$ away from O' . Since AO' is fixed, all the points of our cross-section are the same distance from O' . Therefore, the cross-section must be a circle. (Make sure you see why every point on this circle must be in the cross-section.)

□

While the proof of the volume formula for cones requires some very advanced tools, we can find a formula for the area of the curved surface of a cone with techniques we already have.

Problem 15.5: Find a formula for the lateral surface area of a right circular cone with base radius r and slant height l .

Solution for Problem 15.5: Since cutting and unrolling the curved surface was so successful in finding the lateral surface area of a cylinder, we try it with a cone as well. We cut the curved surface of the cone along $\overline{AB}$, where A is the vertex and B is a point on the circumference of the base.



Since every point on the circumference of the cone's base is the same distance from the cone's vertex (the slant height), when we unroll the curved surface, these points will still all be the same distance from the vertex. Hence, our 'unrolled' surface is a sector of a circle as shown at right above. (B and B' coincide when the sector is rolled up to form a cone.)

The radius of this sector is AB , the slant height of the cone. To find the area of the sector, we must determine what portion of a whole circle the sector is. We know that the length of $\widehat{BB'}$ is just equal to the circumference of the cone's base. The base of the cone has radius r , so its circumference is $2\pi r$. Thus, the length of $\widehat{BB'}$ is $2\pi r$. Since a whole circle with radius $AB = l$ has circumference $2\pi l$, our sector is $(2\pi r)/(2\pi l) = r/l$ of a whole circle.

A full circle with radius l has area πl^2 , so the area of a sector that is r/l of this circle is $(r/l)(\pi l^2) = \pi r l$.

Recall from Problem 14.9 that we showed that the lateral surface area of a regular pyramid is half the product of the slant height and the perimeter of the base. The proof we used there wouldn't work for cones, since we don't have triangular faces as the sides of a cone. However, since cones are essentially just pyramids with circular bases, we expect the formula to work for cones, too. Trying it, we note that the perimeter of the base of a cone is $2\pi r$, so our formula gives us $(1/2)(2\pi r)(l) = \pi r l$ for the lateral surface area. Unsurprisingly, this matches the formula we already proved. □

Important: The lateral surface area of a right circular cone with radius r and slant height l is $\pi r l$.



Concept: As with cylinders, problems involving the curved surface of a cone can often be solved by 'unrolling' the curved surface. As we've seen, the result is a sector of a circle.



Problem 15.6: Find the volume of a right circular cone that has radius 6 and slant height 8.

Solution for Problem 15.6: We need the height in order to find the volume. As we did with a pyramid, we can build a right triangle to find the height. We draw $\overline{AO}$, the axis of the cone, and connect A and O to B , a point on the circumference of the base. Since $\overline{OB}$ is a radius and $\overline{AB}$ is a slant height, we have $AB = 8$ and $OB = 6$. Since the cone is 'right,' $\angle AOB$ is a right angle, so $AO = \sqrt{AB^2 - OB^2} = 2\sqrt{7}$. Therefore, we have:

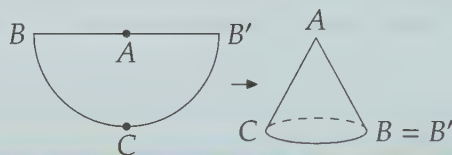
$$\text{Volume} = \frac{\pi r^2 h}{3} = \frac{\pi(6^2)(2\sqrt{7})}{3} = 24\pi\sqrt{7}.$$

□

As we've just seen, a radius, an altitude, and a slant height make a right triangle in a right circular cone. We could immortalize this in yet another formula, $l^2 = r^2 + h^2$, but there's no need to – this is just the Pythagorean Theorem!

Back in Problem 15.5, we unrolled the curved surface of a cone to get a sector. What happens if we go the other way?

Problem 15.7: A semicircle is rolled up to form a cone as shown. If the radius of the semicircle is 8, what is the volume of the cone?



Solution for Problem 15.7: Here we need to find both the radius and the height of the cone. At first it seems that all we are given is the slant height; however, the information about the 'unrolled' curved surface can be used to find the radius in a couple different ways.

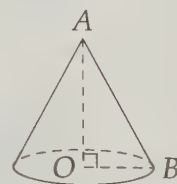
Method One: Roll it up! When we roll up the semicircle into a cone, the arc $\widehat{BCB'}$ becomes the circumference of the base of the cone. Since this arc of the semicircle is half the circumference of a circle with radius 8, it has length 8π . Therefore, the circumference of the base of the cone is 8π , so its radius must be 4.

Method Two: Use the lateral surface area! The semicircle has area $\pi(8^2)/2 = 32\pi$. When rolled up, this is the curved surface of the cone, which we know has area πrl , where r is the radius of the cone and l is its slant height. Since the radius of the semicircle is the slant height of the cone, we have $l = 8$. Hence, we can solve $\pi rl = 32\pi$ to find $r = 32/l = 4$.

Once we have the radius, we can find the height by using right triangle $\triangle AOB$ in the cone as before. Specifically, $h = \sqrt{8^2 - 4^2} = 4\sqrt{3}$. Thus, our volume is

$$\frac{\pi r^2 h}{3} = \frac{64\pi\sqrt{3}}{3}.$$

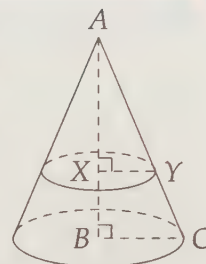
□



Now that we have a good understanding of cone volumes and basic cross-sections of a cone, let's try putting them together.

Problem 15.8: A cone with vertex A , height $AB = 9$, and radius $BC = 12$ is given. The cone is cut in two pieces by a plane perpendicular to $\overline{AB}$ at point X , where $AX = 6$. Find the volume of the two smaller pieces thus formed.

Solution for Problem 15.8: We showed in Problem 15.4 that a cross-section of a cone perpendicular to its axis is a circle. So, one of our pieces is itself a cone. The other piece is called a **right circular frustum**. We don't have any tools to deal with a frustum, but we do know how to find the volume of a cone. The original cone has volume $\pi r^2 h / 3 = \pi(12^2)(9)/3 = 432\pi$. We have the height of the smaller cone, $AX = 6$, so all we have to do is find the radius.



Since $\overline{XY}$ and $\overline{BC}$ are each perpendicular to $\overline{AB}$, we have $\angle AXY = \angle ABC$ and $\angle XAY = \angle BAC$, so $\triangle AXY \sim \triangle ABC$. (Notice that we are essentially considering the cross-section of the cone that contains $\triangle ABC$ here – three-dimensional problems are often just two-dimensional problems in disguise!) Therefore, $XY/AX = BC/AB$, so $XY = (BC/AB)(AX) = (12/9)(6) = 8$. Hence, our little cone has volume $\pi(XY^2)(AX)/3 = 128\pi$.

To get the volume of the other piece, we merely subtract the little cone from the big one, which yields $432\pi - 128\pi = 304\pi$.

Notice that

$$\frac{\text{Volume of small cone}}{\text{Volume of large cone}} = \frac{128\pi}{432\pi} = \frac{8}{27} = \left(\frac{2}{3}\right)^3 = \left(\frac{AX}{AB}\right)^3.$$

This shouldn't be a surprise, because our cones are similar figures. $\square$

Important:



Just as the ratio of the areas of similar two-dimensional figures is the square of the ratio of their corresponding sides, the ratio of the surface areas of similar three-dimensional figures is the square of the ratio of their corresponding side lengths. Moreover, the ratio of the volumes of similar three-dimensional figures is the cube of the ratio of their corresponding side lengths.

Exercises

15.2.1 A cone has height 5 and radius 2. Find the volume and total surface area of the cone.

15.2.2 A cone has a lateral surface area 54π and radius 6. Find the slant height, height, and volume of the cone.

15.2.3 A quarter-circle with radius 4 is rolled up to form a cone.

- (a) Find the lateral surface area of the cone.
- (b) Find the volume of the cone.

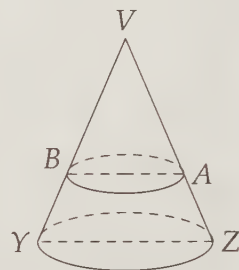
15.2.4 Cone $\mathcal{A}$ has twice the radius, but half the height, of cone $\mathcal{B}$. What is the ratio of the volume of $\mathcal{A}$ to the volume of $\mathcal{B}$?

15.2.5 Prove that the vertex of a right circular cone is equidistant from every point on the circumference of the cone's base. **Hints:** 310

15.2.6 Is it possible for the slant height of a cone to be smaller than the cone's radius?

15.2.7 How could we have used the fact that the two cones in Problem 15.8 are similar in order to find the volume of the small cone quickly?

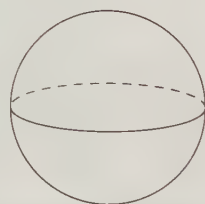
15.2.8★ In this problem we will develop a formula for the volume of a right circular frustum, which was described in Problem 15.8. Call the bases $\mathcal{B}_1$ and $\mathcal{B}_2$ and let their radii be r_1 and r_2 , respectively, with $r_1 < r_2$. Furthermore, let the distance between $\mathcal{B}_1$ and $\mathcal{B}_2$ be h . In Problem 15.8, we formed a frustum by cutting a cone with a plane parallel to the base. Let's try running this backwards – recreating a cone by extending the curved surface of the frustum up to a point. Call this point V , and let $\overline{AB}$ be a diameter of the small base and $\overline{YZ}$ be a diameter of the large base as shown.



- What do we know about $\triangle VAB$ and $\triangle VZY$?
- Use part (a) to find the distance from V to $\mathcal{B}_1$ in terms of h , r_1 , and r_2 .
- What is the volume of the cone with vertex V and base $\mathcal{B}_1$ in terms of h , r_1 , and r_2 ?
- What is the volume of the cone with vertex V and base $\mathcal{B}_2$ in terms of h , r_1 , and r_2 ?
- What is the volume of the frustum in terms of h , r_1 , and r_2 ?

15.3 Spheres

Just as a circle is the set of all points in a plane that are the same distance from a given point, a **sphere** is the set of all points in space that are equidistant from a given point. Most balls and globes are examples of spheres.



Fortunately, we can take a break from the vocabulary bombardment in this chapter. The only significant length we have in a sphere is the sphere's radius, which is just the distance from the center of the sphere to the surface of the sphere. The formulas for the volume and the surface area of a sphere are very challenging to derive (we won't derive them in this book; see page 407 for a hint as to how Archimedes did it). Archimedes considered his discovery of the formula for the volume of a sphere the greatest of all his accomplishments, and even requested a reference to it be inscribed on his tombstone.

Important: A sphere of radius r has:



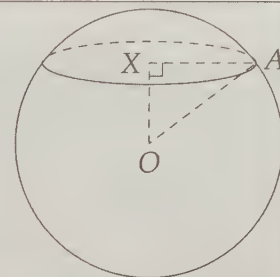
$$\text{Volume} = \frac{4\pi r^3}{3}$$

$$\text{Surface Area} = 4\pi r^2$$

Problems

Problem 15.9: The volume of a sphere is numerically equal to twice its surface area. What is the radius of the sphere?

Problem 15.10: In this problem, we show that if a plane passes through the interior of a sphere, then the intersection of the plane and the sphere is a circle. Let O be the center of the sphere, X be the foot of the perpendicular segment from O to a plane cutting the sphere, and A be a point on the intersection of the plane and the surface of the sphere (as shown).



- What kind of triangle is $\triangle OAX$?
- Find XA in terms of OA and OX .
- Why does your answer to part (b) tell you that the intersection of the plane and the sphere is a circle?

Problem 15.11: A plane cuts through a 26 cm diameter sphere, but the closest it gets to the center is 5 cm. What is the area of the intersection of the sphere and the plane?

Problem 15.12: A sphere is inscribed in a cube, meaning the sphere is tangent to all 6 faces of the cube. A second cube is then inscribed in the sphere, meaning that all 8 vertices of this second cube are on the surface of the sphere. Given that the radius of the sphere is 6, we will find the volume of each cube.

- Forget the small cube for a minute. Draw a cross-section of the large cube with the sphere inside that includes the center of the sphere and the four of the points where the sphere is tangent to the cube.
- Use your cross-section to find the length of an edge of the cube. Find the volume of the cube.
- Forget the large cube for a minute. Draw a cross-section of the sphere and small cube that includes four of the cube's vertices and the center of the sphere. What part of the small cube equals the diameter of the sphere?
- Use the radius of the sphere to find the length of an edge of the small cube, and then find the cube's volume.

We'll start our work with spheres by using the formulas for volume and surface area.

Problem 15.9: The volume of a sphere is numerically equal to twice its surface area. What is the radius of the sphere?

Solution for Problem 15.9: We let the radius of the sphere be r and convert the words in the problem to an equation we can solve: $\text{Volume} = 2 \times (\text{Surface Area})$.

Extra!



$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots$$

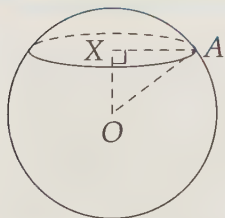
Therefore, we have

$$\frac{4\pi r^3}{3} = 2(4\pi r^2).$$

Solving this equation gives $r = 6$. $\square$

We'll continue our exploration of spheres by considering a cross-section of a sphere.

Problem 15.10: Prove that if a plane passes through the interior of a sphere, then the intersection of the plane and the sphere is a circle.



Solution for Problem 15.10: Intuitively, the cross-section seems to be a circle. We'll prove it is a circle the same way we proved similar cross-sections are circles earlier. Let O be the center of our sphere, r be its radius, and $\mathcal{P}$ be our plane. Let X be the foot of the perpendicular segment from O to $\mathcal{P}$. X is the point we think is the center of our circle. Let A be a point where the surface of the sphere meets $\mathcal{P}$.

Since $\triangle OXA$ is a right triangle, we have $XA = \sqrt{OA^2 - OX^2}$. OA is just the radius of the sphere, so we have $XA = \sqrt{r^2 - OX^2}$. Similarly, every point where the surface of the sphere meets $\mathcal{P}$ is the same distance, $\sqrt{r^2 - OX^2}$, from X . Hence, the intersection of the plane and the sphere is a circle in plane $\mathcal{P}$ with center X and radius $\sqrt{r^2 - OX^2}$. Furthermore, every point in $\mathcal{P}$ that is on the circle with center X and radius XA is on the sphere. (Make sure you see why this is true, and why it is important!)

$\square$

We have now proved two very useful facts about cross-sections of a sphere.

Important:



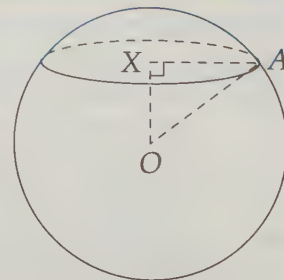
Every cross-section of a sphere is a circle (or a point, when the cross-section plane is tangent to the sphere). The segment connecting the center of the sphere to the center of this circle is perpendicular to the plane of the cross-section.

A cross-section of a sphere that has the center of the sphere as its center is sometimes called a **great circle** of the sphere.

Let's try using our new cross-section knowledge on a problem.

Problem 15.11: A plane cuts through a 26 cm diameter sphere, but the closest it gets to the center is 5 cm. What is the area of the intersection of the sphere and the plane?

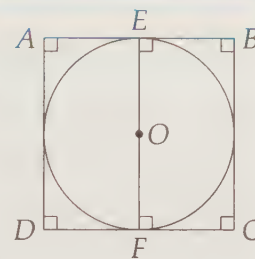
Solution for Problem 15.11: In this problem we are simply considering a specific cross-section of a specific sphere. It's so close to the last problem that we can even use the same diagram. We are given that $AO = 26/2 = 13$ and $OX = 5$. From right triangle $\triangle AXO$ we find $AX = \sqrt{13^2 - 5^2} = 12$ (maybe you recognized the 5-12-13 Pythagorean triple). We know that this cross-section is a circle, so its area is $12^2\pi = 144\pi \text{ cm}^2$. $\square$



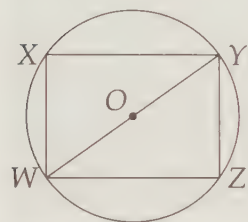
Now we'll use cross-sections to examine a problem that includes more than just spheres.

Problem 15.12: A sphere is inscribed in a cube, meaning the sphere is tangent to all 6 faces of the cube. A second cube is then inscribed in the sphere, meaning that all 8 vertices of this second cube are on the surface of the sphere. Given that the radius of the sphere is 6, find the volume of each cube.

Solution for Problem 15.12: To find the edge length of the larger cube, we consider a cross-section that includes the center of the sphere and four of the points of tangency where the sphere touches the cube at the centers of the faces of the cube. The cross-section of the sphere is a circle. Since the circle includes the center of the sphere, the radius of the circle equals the radius of the sphere. The cross-section of the cube is a square whose side lengths equal the edge lengths of the cube.



Drawing radii to the points of tangency at E and F , we see that $AEFD$ is a rectangle and $\overline{EF}$ is a diameter of the circle. Hence, $AD = EF = 2(OE) = 12$, and the volume of the large cube is $12^3 = 1728$. (Note that we assumed the 'obvious' fact that $\overline{EF}$ passes through O . For an extra challenge, prove it!)



We have several ways to find the volume of the small cube. We'll start with a judiciously chosen cross-section. We want to include the center of the sphere since we know the radius of the sphere. Therefore, we include the center of the sphere and four vertices of the cube as shown at left. $\overline{XY}$ and $\overline{WZ}$ are diagonals of faces of the cube, while $\overline{WX}$ and $\overline{YZ}$ are edges of the cube. Therefore W and Y are opposite vertices of the cube, so that $\overline{WY}$ is a space diagonal of the cube. Since $\overline{WY}$ is also a diameter of the sphere, we know that $WY = 12$. Since $\overline{WY}$ is a space diagonal of the small cube, the side length of the small cube is $WY/\sqrt{3} = 4\sqrt{3}$. So, the volume of the small cube is $(4\sqrt{3})^3 = 192\sqrt{3}$. $\square$

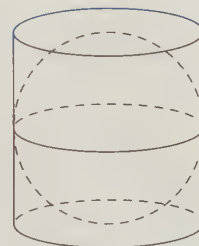
Concept: Cross-sections including the center of the sphere are often useful in problems involving a sphere. Typically, we want to include other important points in the cross-section as well, such as points where the sphere is tangent to other figures in the problem.

Exercises

15.3.1 The volume of a sphere is 36π . What is the surface area of the sphere?

15.3.2 A sphere is inscribed in a cylinder, meaning that it is tangent to both bases, and that one great circle of the sphere is along the curved surface of the cylinder.

- Find the ratio of the volume of the sphere to the volume of the cylinder. **Hints:** 354
- Find the ratio of the surface area of the sphere to the lateral surface area of the cylinder.



15.3.3 An ice cream cone has radius 1 inch and height 4 inches. What is the number of inches in the radius of a sphere of ice cream that has the same volume as the cone? (Source: MATHCOUNTS)

15.3.4 At the end of our solution to Problem 15.10, we noted that ‘Every point in $\mathcal{P}$ that is on the circle with center X and radius XA is on the sphere.’ Why is this true and why is our solution incomplete without this?

15.3.5 A ball was floating in a lake when the lake froze. The ball was removed (without breaking the ice), leaving a hole 24 cm across and 8 cm deep. Find the radius of the ball. (Source: *AHSME*)

15.3.6★ In this problem we explore facts about tangency and spheres.

- Sphere $\mathcal{S}$ is tangent to plane $\mathcal{P}$ at point T , and the center of sphere $\mathcal{S}$ is O . Let X be some point in plane $\mathcal{P}$ besides T . Consider the solution to Problem 12.14 on page 318, then prove that $\overline{OT} \perp \overline{TX}$.
- Spheres $\mathcal{S}_1$ and $\mathcal{S}_2$ are externally tangent at point T . The centers of the spheres are O_1 and O_2 . Consider the solution to Problem 10.17 on page 278, then prove that $\overleftrightarrow{O_1O_2}$ passes through T .
- Suppose $\mathcal{S}_1$ is internally tangent to $\mathcal{S}_2$ at T , meaning $\mathcal{S}_1$ is inside $\mathcal{S}_2$ and both pass through T . Let the centers of $\mathcal{S}_1$ and $\mathcal{S}_2$ be O_1 and O_2 , respectively. Must $\overleftrightarrow{O_1O_2}$ pass through T ?

15.3.7★ A cube is inscribed in a sphere. Prove that each space diagonal is a diameter of the sphere. (Write a complete proof without invoking symmetry. Don’t just say, ‘It’s obvious!’ and move on.) **Hints:** 53, 100, 144

15.4 Problems

The following problems are challenging extensions of the material in this chapter and the previous chapter. Each problem illustrates useful problem solving techniques for three-dimensional geometry problems. Rather than write the terms ‘two-dimensional’ and ‘three-dimensional’ repeatedly, we’ll frequently use ‘2-D’ and ‘3-D’ instead.

Problems

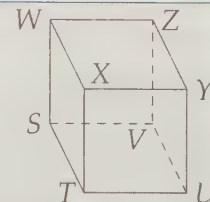
Problem 15.13: My company produces 6 inch tall cans of paint. The cans have a radius of 15 inches. (Yes, these are very oddly shaped cans!) We want to offer a super-size can of paint. Toby wants to leave the can just as wide as it is, but increase the height by x inches. Maryanna thinks the fat cans sell better, so she wants to make them even wider. She wants to increase the radius by x inches. After arguing for a while, they decide to use the design that produces the greatest increase in paint volume.

They reach for their scratch paper and figure out how much more paint is needed to fill their new can designs. They both use the same value of x , and they find that both designs require the *same* increase in paint volume! What value of x did they use?

Problem 15.14: The areas of three of the faces of a rectangular prism are 24, 32, and 36. In this problem we will find the volume of the prism.

- Let the dimensions of the prism be x , y , and z . Use the information given to find three equations.
- In terms of x , y , and z , what is the volume of the prism?
- How can you combine your equations from the first part by adding, subtracting, and/or multiplying them to find the volume of the prism?

Problem 15.15: Cube $STUVWXYZ$ is shown. Find $\angle WYT$.



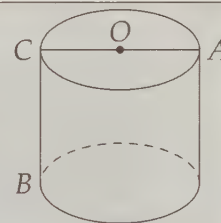
Problem 15.16: A ball with radius 6 inches is tangent to the wall and the floor in my house. Given that the wall is perpendicular to floor, what is the radius of the largest ball I can slide into the space between the ball and the corner where the wall meets the floor? **Hints:** 495, 175

Problem 15.17: Three congruent spheres are inside a cylinder such that each sphere is tangent to the other two spheres, to both bases of the cylinder, and to the curved surface of the cylinder. In this problem we will find the volume of the cylinder given that the radius of each sphere is 12.

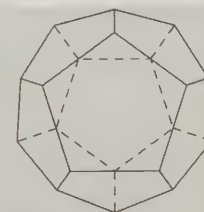
- Find the height of the cylinder.
- Draw a cross-section of the figures in the problem that contains the centers of the spheres. What kind of triangle is formed by connecting the centers of the spheres?
- Let X be the point where your cross-section intersects the axis of the cylinder. How is X related to the triangle mentioned in part (b)?
- Find the radius of the cylinder, then the volume of the cylinder.

Problem 15.18: Adam the Ant is at point A of cube $ABCDEFGH$ with edge length 1. He wishes to walk along the surface of the cube to point G , where G is the vertex of the cube such that $\overline{AG}$ is a space diagonal of the cube. What is the length of the shortest path Adam can take?

Problem 15.19: Adam the Ant is now on point A of the cylinder shown. He wishes to walk along the curved surface of the cylinder to point B , where $\overline{BC}$ is a height of the cylinder and C is on the top base such that $\overline{AC}$ is a diameter as shown. Given that the cylinder has height 4 and radius 3, what is the length of the shortest path Adam can take?



Extra! We saw on page 264 that squares, hexagons, and triangles are the only regular polygons that will tile the plane. Pentagons, with their quirky 108° angles, simply can't add up to 360° , no matter how many of them get together at a vertex. But, on a sphere, pentagons get their due! We can view the dodecahedron we discovered on page 372 (and shown at right) as a tiling of a sphere with regular pentagons.



Each of the other types of polyhedra can be considered a method of tiling a sphere with regular polygons. Notice that while there is only one way to tile a plane with equilateral triangles, there are three ways to tile a sphere with them!

We start off with a couple problems in which we apply algebra to geometry problems. Then we'll move on to challenging spatial geometry problems.

Problem 15.13: My company produces 6 inch tall cans of paint. The cans have a radius of 15 inches. (Yes, these are very oddly shaped cans!) We want to offer a super-size can of paint. Toby wants to leave the can just as wide as it is, but increase the height by x inches. Maryanna thinks the fat cans sell better, so she wants to make them even wider. She wants to increase the radius by x inches. After arguing for a while, they decide to go with the design that produces the greatest increase in paint volume.

They reach for their scratch paper and figure out how much more paint is needed to fill their new can designs. They both use the same value of x , and they find that both designs require the *same* increase in paint volume! What value of x did they use?

Solution for Problem 15.13: We are given

$$\text{Volume(Toby's design)} = \text{Volume(Maryanna's design)},$$

so we find each volume in terms of x :

$$\pi(15^2)(6 + x) = \pi(15 + x)^2(6).$$

After a few lines of algebra, we have $x(2x - 15) = 0$. Since x must be nonzero (otherwise we wouldn't be making much of a change to the can design!), we have $x = 15/2$.

We can quickly check our work by considering the dimensions of the new cylinders. Toby's has radius 15 and height $6 + 15/2 = 27/2$, so it has volume $\pi(15^2)(27/2) = \pi(3^5 \cdot 5^2)/2$. Maryanna's has height 6 and radius $15 + 15/2 = 45/2$, so it has volume $\pi(45/2)^2(6) = \pi(3^5 \cdot 5^2)/2$. (Notice we don't have to bother multiplying all those numbers out. Looking at the prime factorizations allows us to quickly verify that the volumes are the same!)

So, they must have both been using a value of $15/2 = 7.5$ for x . $\square$

Concept: For straightforward problems involving the volume and/or surface area of basic solids, usually we just turn the words of the problem into equations using volume and area formulas. Then we solve the equations, check our work, and make sure we answer the question that is asked.

Problem 15.14: The areas of three of the faces of a rectangular prism are 24, 32, and 36. Find the volume of the prism.

Solution for Problem 15.14: We start off by assigning variables and writing the information in the problem as equations. We let the dimensions of the prism be x , y , and z , so the given information about areas of faces becomes:

$$xy = 24$$

$$xz = 32$$

$$yz = 36$$

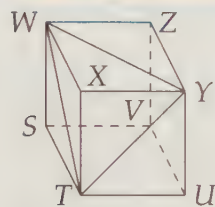
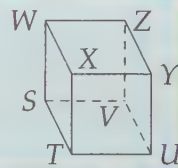
We seek the volume, which in terms of x , y , and z is just xyz . We notice from the nicely symmetric form of the left-hand sides of our equations (xy , yz , zx) that if we multiply all three, we'll get $x^2y^2z^2$. Specifically, we find that multiplying all three equations gives $x^2y^2z^2 = (24)(32)(36)$. Taking the square root of this yields $xyz = \sqrt{(24)(32)(36)} = 96\sqrt{3}$, which is the volume of the prism.

Notice that we didn't even find x , y and z ! But now that we have xyz , we can find them quickly. You'll get a chance to do so as an Exercise. $\square$

Concept: Sometimes turning a geometry problem into algebra results in a nice symmetric system of equations, as we saw in Problem 15.14. We often solve such symmetric systems by adding or multiplying all the given equations.

Now we turn to using our two-dimensional problem solving strategies on three-dimensional problems.

Problem 15.15: Cube $STUVWXYZ$ is shown. Find $\angle WYT$.



Solution for Problem 15.15: We don't really know how to handle angles in three dimensions, so we consider a two-dimensional figure that includes our angle. In other words, we consider the cross-section that contains $\angle WYT$. We form this cross-section by cutting through vertices W , Y , and T of the cube, as shown. Thus, our cross-section is a triangle. Moreover, we note that since each of the sides of $\triangle WYT$ is a diagonal of a face of the cube, the sides of $\triangle WYT$ are all equal. Hence, $\triangle WYT$ is equilateral, so each of its angles, including $\angle WYT$, is 60° . $\square$

Concept: Most 3-D problems are 2-D problems in disguise. Considering cross-sections that include important pieces of the problem often allows us to use our two-dimensional strategies to find the solution.

We'll try a couple more 3-D problems that can be solved with well-chosen cross-sections.

Problem 15.16: A ball with radius 6 inches is tangent to the wall and the floor in my house. Given that the wall is perpendicular to floor, what is the radius of the largest ball I can slide into the space between the ball and the corner where the wall meets the floor?

Extra! The mathematician Johannes Kepler wondered how densely spheres can fill space. The typical stacking you see for oranges at the grocery store fills just 74% of space. Is there a different arrangement that gets more oranges in the same space? It took nearly 400 years before mathematicians Thomas Hales and Samuel Ferguson were able to answer Kepler's question and prove in 1998 what grocers have known all along: there isn't a better way to pack oranges.

Solution for Problem 15.16: We have a 3-D problem involving spheres that are tangent to each other and to some planes, so we consider a cross-section including the centers of the spheres and points of tangency. Hopefully this allows us to build helpful right triangles. (Tangents mean right angles!)

O and P are the centers of our spheres, which are tangent at X . These spheres are tangent to the floor at M and N as shown. Sphere O is tangent to the wall at L . Since the spheres are tangent to the wall and the floor, O and P are equidistant from both. Therefore, both O and P are on the angle bisector of $\angle LCM$ as shown. Since $\angle LCM = 90^\circ$, we have $\angle OCM = \angle PCN = 45^\circ$. So, $\triangle OCM$ is an isosceles right triangle. Similarly, $\triangle PCN$ is an isosceles right triangle, so $PC = PN \sqrt{2} = 6\sqrt{2}$, and $XC = PC - PX = 6\sqrt{2} - 6$. If we let the radius of our small sphere be r , we now know that

$$OC = XC - r = 6\sqrt{2} - 6 - r.$$

However, from $\triangle OCM$, we also have

$$OC = OM \sqrt{2} = r \sqrt{2}.$$

Therefore, we have two expressions for OC , which we can set equal:

$$r \sqrt{2} = 6\sqrt{2} - 6 - r.$$

Therefore, $r + r \sqrt{2} = 6\sqrt{2} - 6$, so

$$r = \frac{6\sqrt{2} - 6}{1 + \sqrt{2}}.$$

We can rationalize the denominator by multiplying the top and bottom of the fraction by $1 - \sqrt{2}$:

$$r = \frac{6\sqrt{2} - 6}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} = \frac{-18 + 12\sqrt{2}}{-1} = 18 - 12\sqrt{2}.$$

□

Concept:



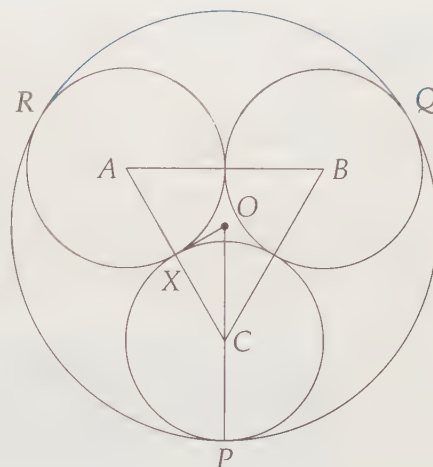
A great many geometry problems are solved by assigning a variable to a length, then finding two different expressions for some other length in terms of that variable. We can then set these expressions equal to solve for the variable.

Problem 15.17: Three congruent spheres are inside a cylinder such that each sphere is tangent to the other two spheres, to both bases of the cylinder, and to the curved surface of the cylinder. Find the volume of the cylinder given that the radius of each sphere is 12.

Solution for Problem 15.17: Since each sphere is tangent to the top and bottom base of the cylinder (and these bases are parallel), the height of the cylinder must equal the diameter of the sphere. Thus, the height of the cylinder is $2(12) = 24$.

To get the radius of the cylinder, we start with a well-chosen cross-section. We have spheres and tangency, so we take the cross-section that includes the centers of the three spheres, the points where the spheres touch each other, and the points where spheres touch the cylinders. Our cross-section is shown at right. $\triangle ABC$ connects the centers of our spheres, and O is where the cross-section meets the axis of the cylinder.

O is equidistant from P , Q and R because the cross-section of the cylinder is a circle with center O . Since $\odot A$, $\odot B$, and $\odot C$ are tangent to $\odot O$, rays $\overrightarrow{RA}$, $\overrightarrow{QB}$, and $\overrightarrow{PC}$ all pass through O . O is therefore equidistant from A , B , and C because $OR = OQ = OP$ and $AR = BQ = CP$. So, O is the circumcenter of $\triangle ABC$. Since $OP = OC + CP$, we only need to find OC to find the radius of the cylinder.



$\triangle ABC$ is equilateral because each side equals twice the radius of a sphere. Therefore, O is also the incenter and the centroid of $\triangle ABC$. Moreover, the points where the spheres' cross-sections touch each other are midpoints of the sides of $\triangle ABC$. Hence, $\triangle OXC$ is a 30-60-90 right triangle because $\overline{OC}$ bisects $\angle ACB$ and $\overline{OX} \perp \overline{AC}$ (make sure you see why). Therefore,

$$OC = 2OX = \frac{2XC}{\sqrt{3}} = \frac{24}{\sqrt{3}} = 8\sqrt{3}.$$

The radius of the cylinder is then $OC + CP = 8\sqrt{3} + 12$. Thus, our volume is:

$$\pi r^2 h = 24\pi(8\sqrt{3} + 12)^2 = 24\pi(4^2)(2\sqrt{3} + 3)^2 = (24 \cdot 16)\pi(21 + 12\sqrt{3}) = 8064\pi + 4608\pi\sqrt{3}.$$

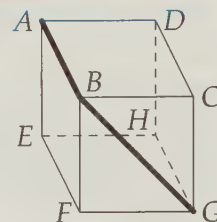
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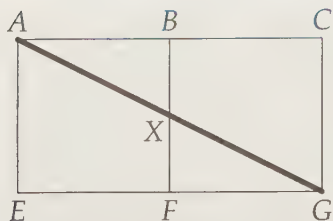
Concept: To solve 3-D problems involving a sphere, consider cross-sections including the center of the sphere. If you have multiple spheres in the problem, try finding a cross-section that includes multiple centers. If the sphere is tangent to anything, try including the point of tangency.

Sometimes the 2-D problem is more subtly hidden in our 3-D problem. In these cases we need more than a simple cross-section to coax it out.

Problem 15.18: Adam the Ant is at point A of cube $ABCDEFGH$ with edge length 1. He wishes to walk along the surface of the cube to point G , where G is the vertex of the cube such that $\overline{AG}$ is a space diagonal of the cube. What is the length of the shortest path Adam can take?

Solution for Problem 15.18: We might think that Adam should walk along edge $\overline{AB}$ and then along face diagonal $\overline{BG}$. However, if we put a big glob of jelly at G and put ants at A , the ants will quickly find a shorter path. They might go diagonally across face $ABFE$ to a point X on $\overline{BF}$, then on to point G . We suspect point X will be the midpoint of $\overline{BF}$, but how can we be sure? (And worse yet, what if $ABCDEFGH$ isn't a cube? You'll get a shot at dealing with that complication in the Exercises, of course!)

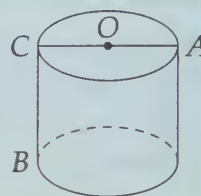




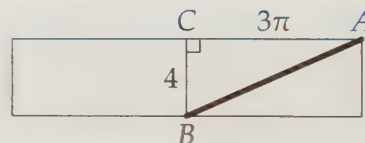
We make our 3-D problem into a 2-D problem by unfolding the cube and placing the two faces Adam crawls across in the same plane as shown. This doesn't change the distance Adam must crawl at all, but it does make our answer clear. Adam's shortest path is the straight segment from A to G. Since $AB = BC = CG = 1$, we have $AG = \sqrt{CG^2 + AC^2} = \sqrt{5}$. Therefore, the shortest path Adam can take has length $\sqrt{5}$. Notice that since $AB = FG$ and $\overline{AC} \parallel \overline{EG}$, we can use congruent triangles $\triangle ABX \cong \triangle GFX$ (by ASA) to show that $BX = FX$, i.e., that Adam's shortest path indeed goes through the midpoint of one of the edges of the cube. $\square$

Concept: The 2-D problem lurking in a 3-D problem isn't always just a cross-section. Sometimes unfolding or unrolling a 3-D figure reveals a simpler 2-D problem.

Problem 15.19: Adam the Ant is now on point A of the cylinder shown. He wishes to walk along the curved surface of the cylinder to point B, where $\overline{BC}$ is a height of the cylinder and C is on the top base such that $\overline{AC}$ is a diameter as shown. Given that the cylinder has height 4 and radius 3, what is the length of the shortest path Adam can take?



Solution for Problem 15.19: When we found the lateral surface area of a cylinder, we discovered how to deal with problems involving the curved surface of a cylinder – we unroll the cylinder. We try that here and we get the rectangle shown at right.



BC still equals the height of the cylinder, so $BC = 4$. However, here $\overline{AC}$ is not a diameter of the original cylinder, but rather the distance from A to C along the circumference of the base. Therefore, AC is half the circumference of the base of our cylinder, or 3π . From right triangle $\triangle ABC$ we can now find $AB = \sqrt{AC^2 + BC^2} = \sqrt{16 + 9\pi^2}$. (Note that this is *not* the distance from A to B in space in our original cylinder. This is the length from A to B along the curved surface of the cylinder. Also, note that if Adam is willing to walk across a base of the cylinder, there is a shorter path he can take!) $\square$

Exercises

15.4.1 The areas of three of the faces of a rectangular prism are 24, 32, and 36. Find the dimensions of the prism. **Hints:** 186

15.4.2 $ABCD$ is a regular tetrahedron. G is the centroid of $\triangle ABC$ and H is the centroid of $\triangle BCD$. Given $AB = 8$, find GH . **Hints:** 198

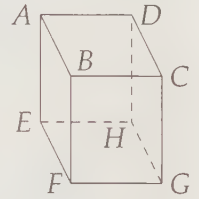
15.4.3 $WXYZ$ is a regular tetrahedron such that $WX = 4$. M is the midpoint of $\overline{WX}$ and N is the midpoint of $\overline{YZ}$.

(a) Prove that $\overline{MN} \perp \overline{WX}$.

(b)★ Find MN .

15.4.4 Three spherical balls are snugly in a row inside a cylindrical can such that the first is tangent to the lid of the can and the middle ball, the second is tangent to the other two balls, and the third is tangent to the bottom of the can and the middle ball. Each ball fits snugly inside the can so that it is tangent to the curved surface of the can and cannot move side-to-side. Given that the radius of each ball is 2 inches, what is the volume of the can?

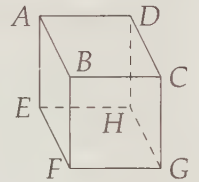
15.4.5 Annie the ant is on vertex A of right rectangular prism $ABCDEFGH$ shown. Given that $AB = 4$, $BC = 6$, and $AE = 8$, what is the shortest distance Annie can walk to reach vertex G ?



15.4.6 Spheres S and T have radii 6 and 8, respectively, and centers O and P , respectively. $OP = 10$.

- Why must the spheres intersect?
- Let X be a point that is on both spheres. What is XO ? What is XP ?
- What kind of triangle is $\triangle XOP$?
- Let the altitude from X to $\overline{OP}$ meet $\overline{OP}$ at Y . What is XY ?
- What is OY ?
- Suppose we pick another point Z that is on both spheres. Must $XY = ZY$?
- Use the previous part to show that the intersection of the two spheres is a circle.
- Find the area of this circle.

15.4.7 $ABCDEFGH$ is a cube as shown. M is the midpoint of $\overline{FG}$ and N is the midpoint of $\overline{GH}$. Given that $AB = 6$, find the following:



- MN .
- AM .
- ★ $[AMN]$. **Hints:** 227
- ★ the volume of $AEMN$. **Hints:** 265

15.4.8 The height of a cylindrical pole is 12 feet and its circumference is 2 feet. A rope is attached to a point on the circumference at the bottom of the pole. The rope is then wrapped tightly around the pole four times before it reaches a point on the top directly above the starting point at the bottom. What is the minimum number of feet in the length of the rope? (*Source: MATHCOUNTS*)

15.4.9★ Two regular square pyramids have all edges 12 cm in length. The pyramids have parallel bases and parallel edges, and each has a vertex at the center of the other pyramid's base. What is the total number of cubic centimeters in the volume of the solid of intersection of the two pyramids? (*Source: MATHCOUNTS*) **Hints:** 318

Extra! We've seen that regular hexagons can tile a plane, and regular pentagons can tile a sphere, giving us a dodecahedron. There's one well-known example of a tiling that uses both hexagons and pentagons – it's commonly known as a soccer ball. This fabulous structure has been around since before the invention of soccer, too, in the form of **buckminsterfullerene**, C_{60} , a recently discovered form of carbon. Drs. Richard Smalley and Robert Curl received the Nobel Prize in 1996 for that discovery.

15.5 Summary

Definitions: If a prism has congruent curved figures as bases instead of polygons, we call it a **cylinder**. When we talk about a cylinder, we almost always mean a **right circular cylinder**, in which the bases are circles. The line connecting the centers of the bases is the **axis** of the cylinder.

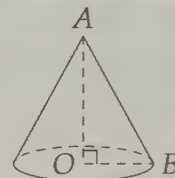
Important: A cylinder with height h and radius r has:



$$\begin{aligned}\text{Volume} &= \pi r^2 h \\ \text{Lateral Surface Area} &= 2\pi r h \\ \text{Total Surface Area} &= 2\pi r h + 2\pi r^2\end{aligned}$$

Don't memorize these formulas! If you take the time to understand them, they'll always be obvious to you.

Definitions: A pyramid with a curved base is a **cone**. At the right is a **right circular cone**, which is usually what we mean when we write 'cone.' The point at the tip of the cone is the **vertex**. The distance from the vertex to the base is the **height** and the line connecting the vertex to the center of the base is the **axis** of the cone. In a right circular cone, the distance from the vertex to a point on the circumference of the base is the **slant height**.



Important: In a cone with radius r , slant height l , and height h , we have:



$$\begin{aligned}\text{Volume} &= \frac{1}{3} \pi r^2 h \\ \text{Lateral Surface Area} &= \pi r l \\ \text{Total Surface Area} &= \pi r l + \pi r^2\end{aligned}$$

Definition: The set of all points in space that are the same distance from a given point is a **sphere**.

Important: A sphere of radius r has:



$$\begin{aligned}\text{Volume} &= \frac{4\pi r^3}{3} \\ \text{Surface Area} &= 4\pi r^2\end{aligned}$$

Every cross-section of a sphere is a circle (or a point, when the plane is tangent to the sphere). The segment connecting the center of the sphere to the center of this circle is perpendicular to the plane of the cross-section.

Important:  Just as the ratio of the areas of similar two-dimensional figures is the square of the ratio of their corresponding sides, the ratio of the surface areas of similar three-dimensional figures is the square of the ratio of their corresponding side lengths. Moreover, the ratio of the volumes of similar three-dimensional figures is the cube of the ratio of their corresponding side lengths.

Problem Solving Strategies

Concepts:



- Problems involving the curved surface of a cylinder can often be solved by ‘unrolling’ the curved surface into a rectangle.
- Similarly, problems involving the curved surface of a cone can often be solved by ‘unrolling’ the curved surface into a sector.
- Basic area/volume word problems are often no different from other word problems. The key to solving them is to assign variables and use the area and volume information to set up equations with the given information.
- To solve 3-D problems involving a sphere, consider cross-sections including the center of the sphere. If you have multiple spheres in the problem, try finding a cross-section that includes multiple centers. If the sphere is tangent to anything, try including the point of tangency.
- Most 3-D problems are 2-D problems in disguise. Usually considering cross-sections including important pieces of the problem allow us to use our two-dimensional strategies to find the solution.
- The most useful of these 2-D strategies is the Pythagorean Theorem. Building right triangles is just as powerful in three dimensions as in two.
- The 2-D problem lurking in a 3-D problem isn’t always just a cross-section. Sometimes we have to manipulate our 3-D figures a little through unfolding or unrolling to discover our 2-D problem.
- A great many geometry problems are solved by assigning a variable to a length, then finding two different expressions for some other length in terms of that variable. We can then set these expressions equal to solve for the variable.

Extra!



$$\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdot \frac{8}{7} \cdot \frac{8}{9} \cdots$$

REVIEW PROBLEMS

- 15.20 The number of cubic inches in a sphere's volume equals the number of square inches in its surface area. What is the radius of the sphere?
- 15.21 A cone has slant height 7 and height 5.
- (a) Find the volume of the cone.
 - (b) Find the total surface area of the cone.
- 15.22 A sphere has diameter 8. Find the surface area and the volume of the sphere.
- 15.23 A right circular cylinder has radius 3 and height 6.
- (a) Find the volume of the cylinder.
 - (b) Find the lateral surface area of the cylinder.
 - (c) Find the total surface area of the cylinder.
- 15.24 A space diagonal of cube C is a diameter of sphere S . The edges of the cube have length 8. Find the volume of the sphere.
- 15.25 The radius of cylinder $\mathcal{A}$ is three times the radius of cone $\mathcal{B}$, but the height of cylinder $\mathcal{A}$ is half the height of cone $\mathcal{B}$. Find the ratio of the volume of $\mathcal{A}$ to the volume of $\mathcal{B}$.
- 15.26 $ABCD$ is a square inscribed in one base of a cylinder, and $EFGH$ is a square inscribed in the other base of the cylinder. Given that $ABCDEFGH$ is a cube with side length 9, find the volume of the cylinder.
- 15.27 A 40° sector of a circle with radius 9 is rolled up to form a cone. Find the volume of the cone.
- 15.28 The intersection of a plane with sphere $\mathcal{G}$ is a circle with area 36π . Find the volume of the sphere if the center of this circle is 8 units from the center of $\mathcal{G}$.
- 15.29 Two spheres are inside a rectangular box such that each sphere is tangent to five faces of the box and to the other sphere. Each sphere has radius 4. Find the volume of the box.
- 15.30 A cross-section of cylinder C is a square in a plane parallel to the axis of C . Given that the area of the square is 36 square units and that the plane of the cross-section is 4 units from the axis of the cylinder, find the volume of the cylinder.
- 15.31 A cone with radius 9 and height 12 is cut in two pieces by a plane parallel to the base of the cone such that the plane is 8 units from the base of the cone. Find the total surface area of each piece thus formed.
- 15.32 Sphere S is tangent to all 12 edges of a cube with edge length 8. Find the volume of the sphere.

Extra! *I don't like that sort of school . . . where the bright childish imagination is utterly discouraged . . . where I have never seen among the pupils, whether boys or girls, anything but little parrots and small calculating machines.*

—Charles Dickens

15.33 A sphere is inscribed in a cone as shown at right. The cone has radius 9 and height 12. Find the radius of the sphere.



15.34 In this problem we use volume to find the radius of a sphere inscribed in regular tetrahedron $ABCD$ with edge length 6. You may want to review Problem 7.14, since we use a similar tactic on this problem.

- Find the length of the altitude from D to face ABC of the tetrahedron.
- Find the volume of tetrahedron $ABCD$.
- Let the radius of the sphere inscribed in the tetrahedron be r and the center of this sphere be O . Why is O a distance of r away from each face of the tetrahedron?
- In terms of r , what is the volume of tetrahedron $OABC$? **Hints:** 402
- In terms of r , what are the volumes of tetrahedra $OABD$, $OACD$, and $OBCD$?
- Find r .

15.35 The height of right circular cone C equals the cone's radius in length. Given that A is the vertex of the cone and $\overline{XY}$ is a diameter of the cone's base, prove that $\angle XAY$ is a right angle.

Challenge Problems

15.36 Right circular cone C has vertex V , radius 6, and height 6. Sphere S has center V and is tangent to the base of cone C .

- What is the radius of the sphere?
- Let X be a point where S meets the curved surface of C . Find the distance from X to the axis of the cone. **Hints:** 11
- Show that the intersection of S and the curved surface of C is a circle.
- Find the area of the circle from the previous part.

15.37 A cylinder is inscribed in a right regular hexagonal prism such that each base of the cylinder is a circle that is tangent to all six sides of a base of the prism. Find the ratio of the volume of the cylinder to the volume of the prism.

15.38 Dennis's scoop of ice cream has a radius of 2 cm. It rests in a cone that has a radius of $\sqrt{3}$ cm at the widest part, and the scoop is tangent to each line containing a slant height of the cone. He eats some of the ice cream and then finds that the remainder of the ice cream can be pushed down to fill the cone exactly. How many cubic centimeters of ice cream did he eat? (Source: Mandelbrot) **Hints:** 177

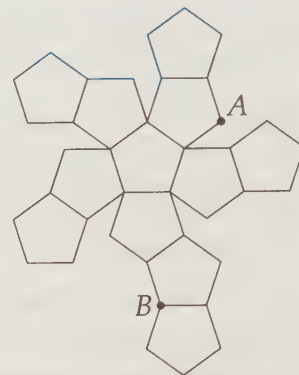


Extra! *Imagination is more important than knowledge.*



—Albert Einstein

15.39 An edgy spider walks along the edges from A to B of the dodecahedron formed by the folding up the figure shown at right. What is the number of edges in the shortest path that spider could take? (Source: MATHCOUNTS)



15.40 Two parallel planes intersect sphere S , forming two congruent circles. Show that the center of the sphere is the midpoint of the segment connecting the centers of the two circles. **Hints:** 489, 556

15.41 Planes $\mathcal{P}_1$ and $\mathcal{P}_2$ are perpendicular. Their intersection is line m . Plane $\mathcal{P}_1$ is 4 units from point O and plane $\mathcal{P}_2$ is 6 units from O . A sphere with radius 9 centered at O meets line m at points A and B . Find AB . **Hints:** 503, 567

15.42 X is on the circumference of one base of a cylinder and Y is on the other base such that $\overline{XY}$ is bisected by the axis of the cylinder. Given that the diameter of the cylinder is 8 and $XY = 12$, find the volume of the cylinder.

15.43 Arnav the ant is on the outside of a cylindrical glass, halfway up the glass. The glass is 8 inches tall and has a radius of 2 inches. Arnav wants to walk to a point on the inside of the glass that is diametrically opposite the point he's now at (i.e., halfway up the glass, on the inside surface of the glass, exactly opposite where he is now). What is the shortest distance he can walk? **Hints:** 218, 529

15.44 A spiral staircase with radius 3 feet turns 270° as it rises 10 feet. What is the number of feet in the length of the outer handrail? (The outer handrail is a curved rail along the entire staircase. It is everywhere 3 feet from the central axis of the staircase.) (Source: MATHCOUNTS) **Hints:** 66

15.45 A right circular cylinder with its diameter equal to its height is inscribed in a right circular cone. The cone has diameter 10 and altitude 12, and the axes of the cylinder and cone coincide. Find the radius of the cylinder. (Source: AMC 10) **Hints:** 429

15.46 $PQRS$ is a regular tetrahedron. The distance from the midpoint of $\overline{PQ}$ to the midpoint of $\overline{RS}$ is 6 units. Find the volume of $PQRS$. **Hints:** 565

15.47 The wire frame at left below consists of three mutually perpendicular segments $\overline{AD}$, $\overline{BD}$, and $\overline{CD}$, each 3 cm in length. A quarter-circle of radius 3 is attached to each pair of segments as shown. The curved portions of the wire frame will fit snugly against a sphere of a certain size, so that the entire lengths of all three quarter circles make contact with the sphere. For what radius is this possible? (Source: Mandelbrot) **Hints:** 389, 452

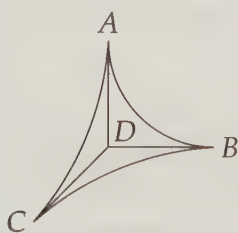


Figure 15.4: Diagram for Problem 15.47

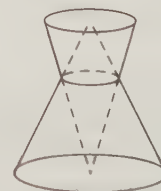


Figure 15.5: Diagram for Problem 15.48

15.48★ One cone of radius 4 and height 12 and another cone of radius 6 and height 12 intersect as shown at right above, so that the vertex of one coincides with the center of the base of the other. Find the volume of the intersection of the two cones. **Hints:** 139, 238

15.49★

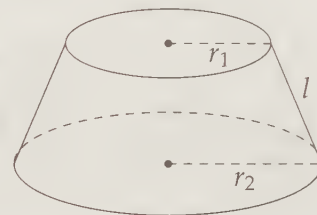
- (a) In tetrahedron $ABCD$, let V be the volume of the tetrahedron and r the radius of the sphere that is tangent to all four faces of the tetrahedron. Let K_1 , K_2 , K_3 , and K_4 denote the areas of faces BCD , ACD , ABD , and ABC , respectively. Prove that

$$r = \frac{3V}{K_1 + K_2 + K_3 + K_4}.$$

Hints: 184

- (b) $WXYZ$ is a tetrahedron with $XY = YZ = ZX = 9$ and $WX = WY = WZ = 18$. Find the radius of the sphere that is inscribed in tetrahedron $WXYZ$. (You might remember working with this pyramid in Problem 14.26.)

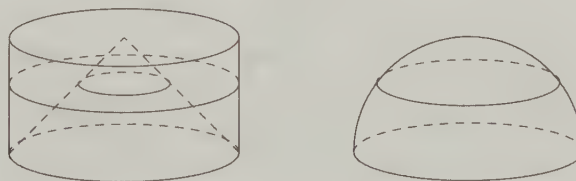
15.50★ Find a formula for the total surface area of a right circular frustum with base radii r_1 and r_2 and slant height l . (The slant height of a frustum is the distance from a point on the circumference of one base to the nearest point on the circumference of the other base, as shown at right.) **Hints:** 219, 292



15.51★ A ball of radius 1 is in the corner of a room, tangent to two walls and the floor. A smaller ball is also in the corner, also tangent to both walls, the floor, and the larger ball. The walls are perpendicular to each other, and they are perpendicular to the floor. Find the radius of the smaller ball. **Hints:** 327

Extra! We've already had a glimpse of the brilliance of **Archimedes** on page 293 in his proof of the formula for the area of a circle. Although the mathematical and scientific accomplishments of Archimedes could fill several books, the feat he allegedly treasured most was his determination of the volume and surface area of a sphere. Legend has it that he even asked that his tombstone be inscribed with a sphere inscribed in a cylinder to commemorate his feat.

Archimedes used the tactics we saw in his area of a circle proof, although with considerably more complicated figures. He compared a sphere to figures he already knew how to handle: cones and cylinders. To get a feel for the volume formula, consider the figures below. Shown is a cone that shares a base with a cylinder such that both the cone and the cylinder have height equal to the radius of the base. Next to these is a **hemisphere**, or half a sphere, with the same radius as the base of the cylinder and cone.



Consider the cross-sections of each taken at the same height, h , from the base. Letting the radius of the hemisphere be r , find the areas of the cross-section of each figure in terms of r and h . Next, find the area inside the cross-section of the cylinder, but outside the cross-section of the cone. Notice anything interesting?